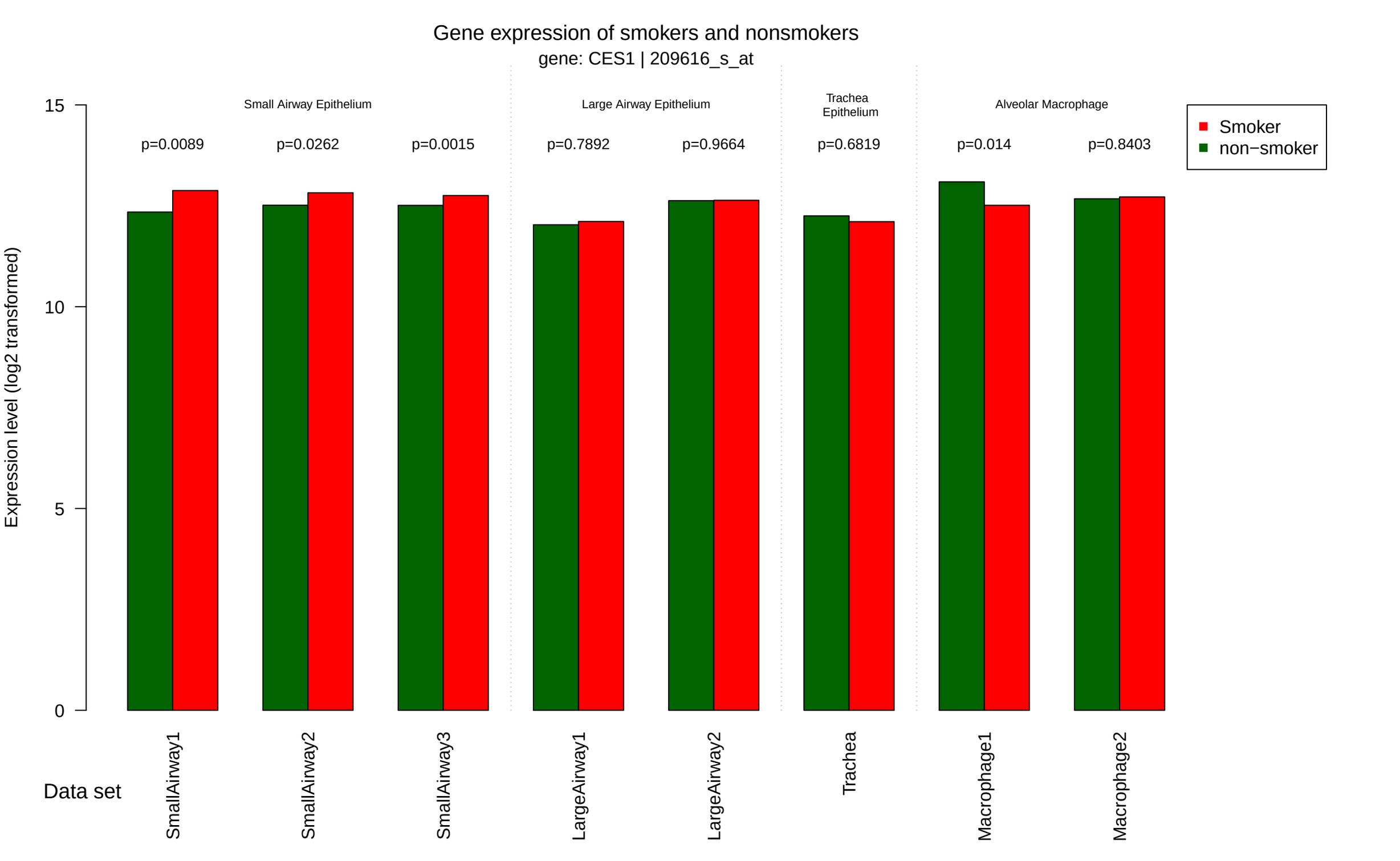
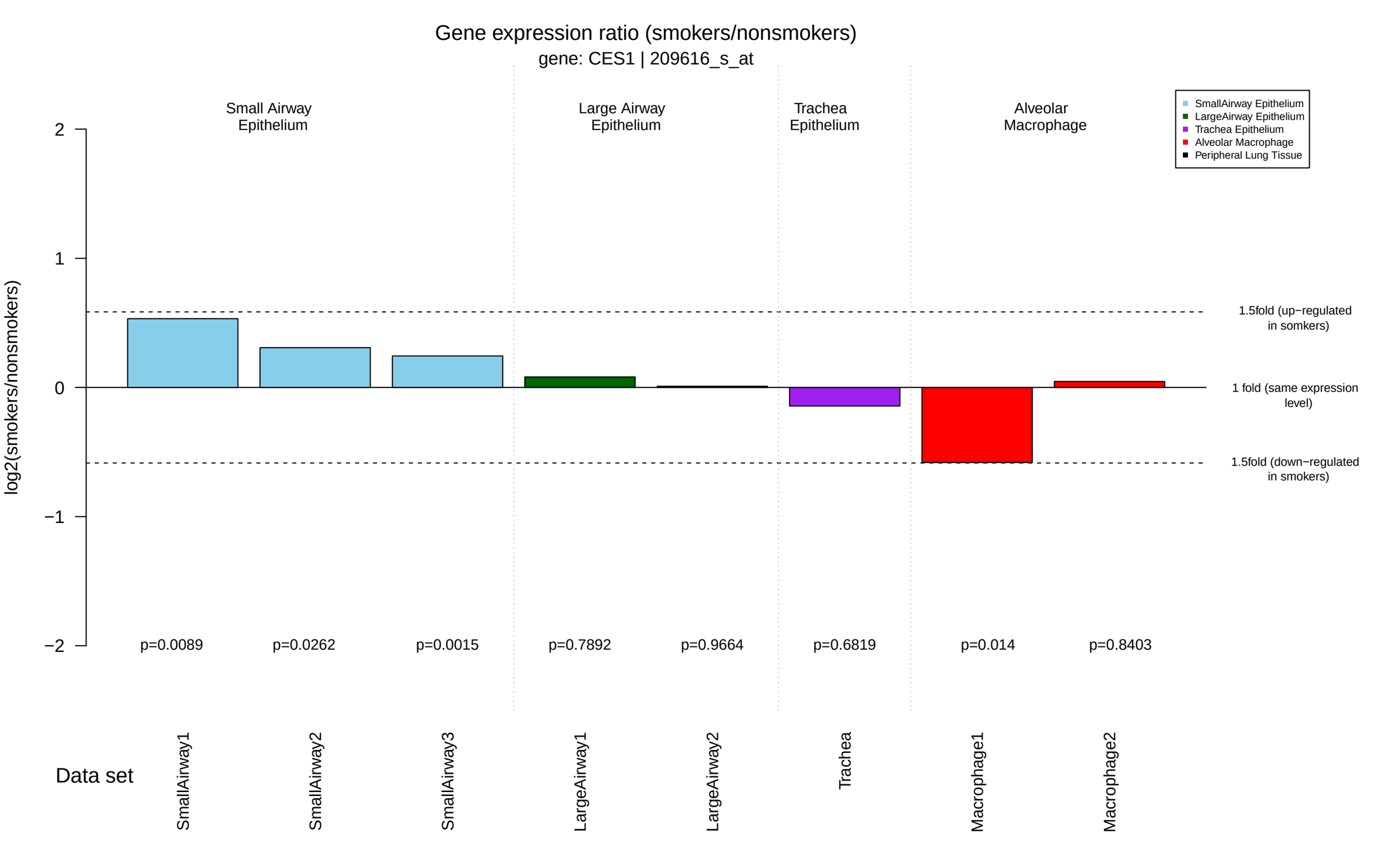
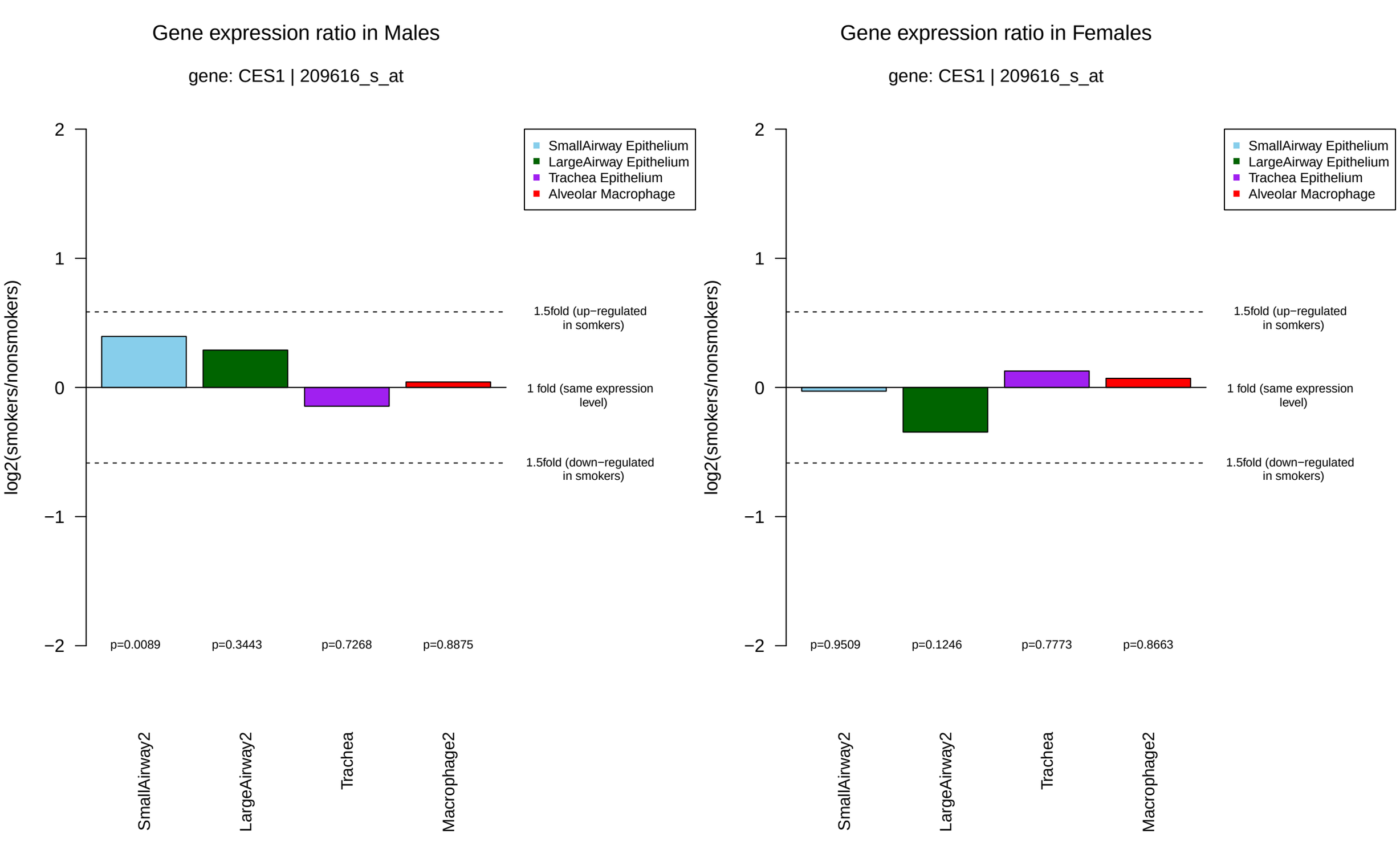
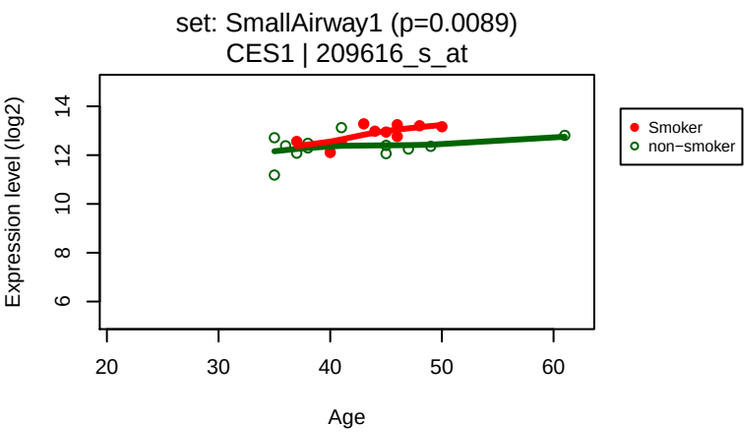
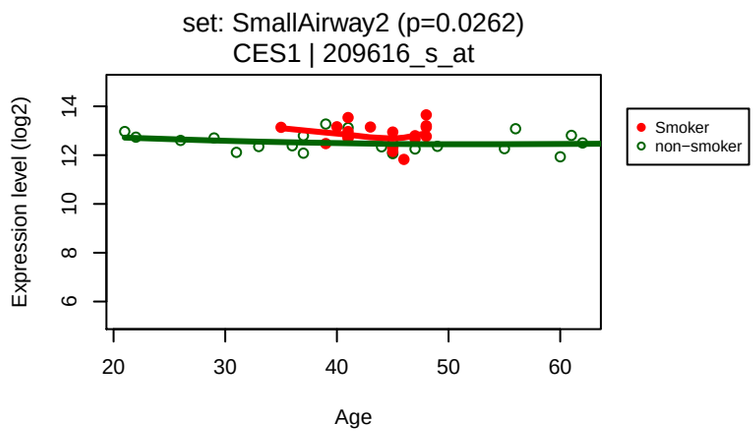
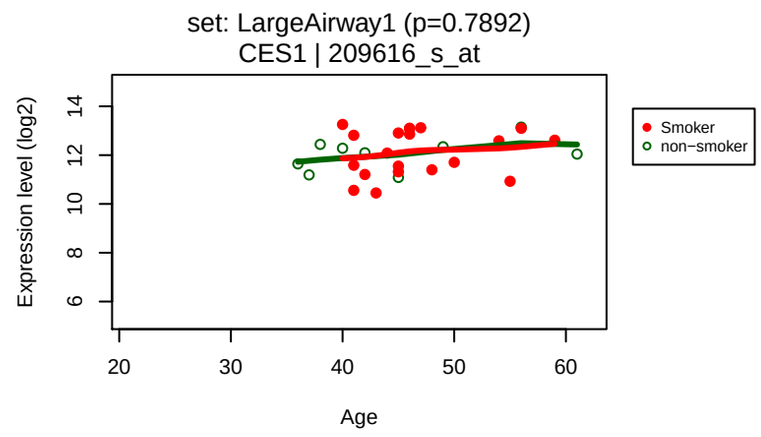

Gene expression of smokers and nonsmokers
gene: CES1 | 209616_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0089
p=0.0262
p=0.0015
p=0.7892
p=0.9664
p=0.6819
p=0.014
p=0.8403
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

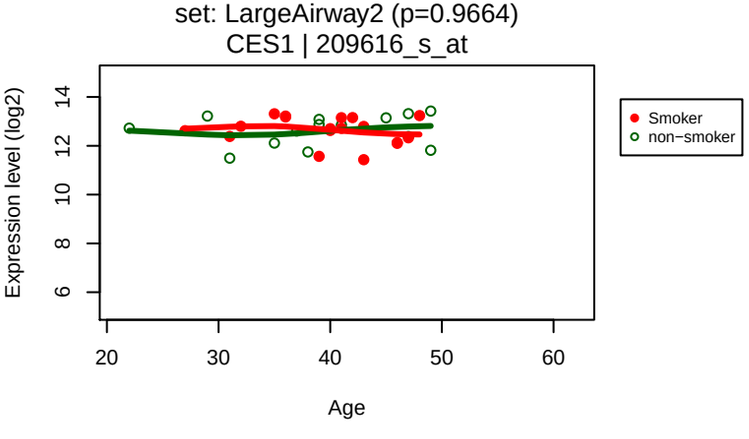

Gene expression ratio (smokers/nonsmokers)
gene: CES1 | 209616_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0089
p=0.0262
p=0.0015
p=0.7892
p=0.9664
p=0.6819
p=0.014
p=0.8403
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

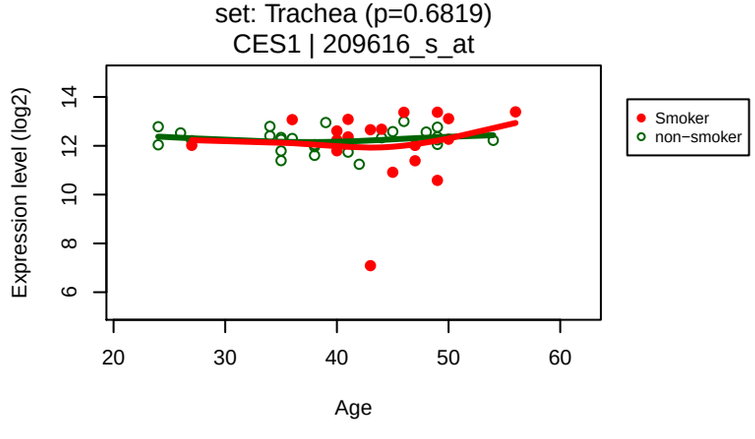

Gene expression ratio in Males
gene: CES1 | 209616_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0089
p=0.3443
p=0.7268
p=0.8875
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CES1 | 209616_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9509
p=0.1246
p=0.7773
p=0.8663
SmallAirway2
LargeAirway2
Trachea
Macrophage2

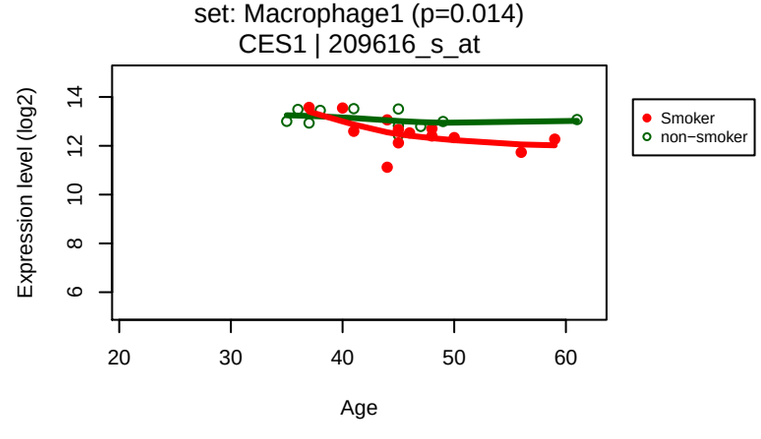

set: SmallAirway1 (p=0.0089)
CES1 | 209616_s_at
Expression level (log2)
Age
Smoker
non−smoker

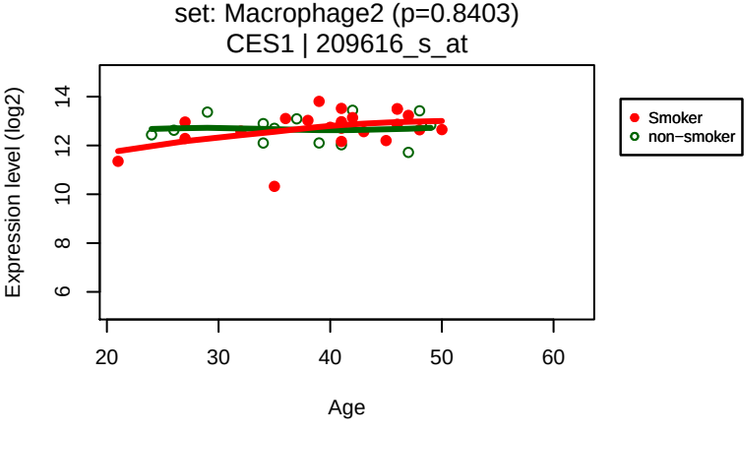

set: SmallAirway2 (p=0.0262)
CES1 | 209616_s_at
Expression level (log2)
Age
Smoker
non−smoker

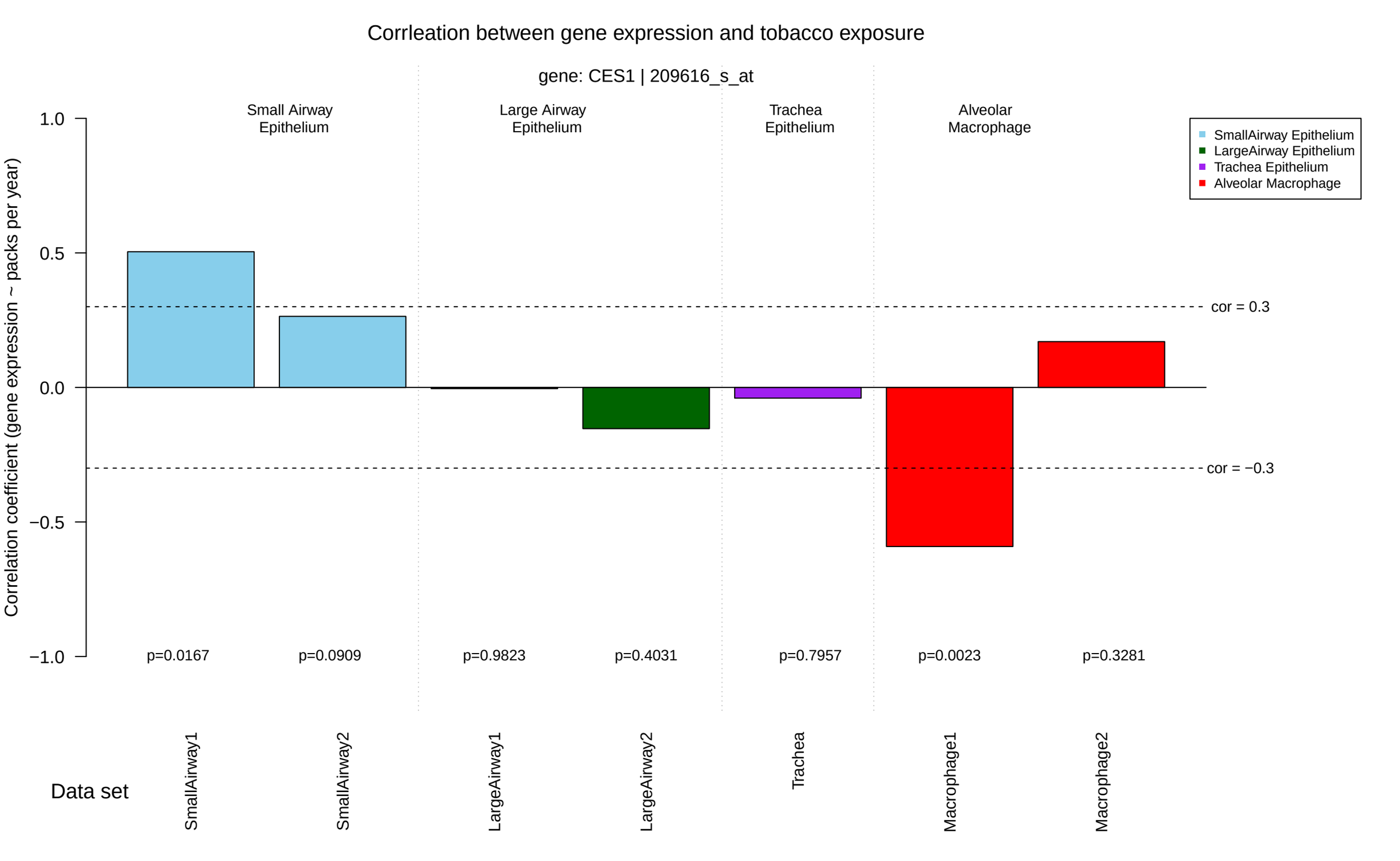

set: LargeAirway1 (p=0.7892)
CES1 | 209616_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9664)
CES1 | 209616_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6819)
CES1 | 209616_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.014)
CES1 | 209616_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8403)
CES1 | 209616_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CES1 | 209616_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0167
p=0.0909
p=0.9823
p=0.4031
p=0.7957
p=0.0023
p=0.3281
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2